TSINGHUA SCIENCE AND TECHNOLOGY
ISSN 1007-0214 pp1–23
DOI: 10.26599/TST.2025.9010146
Volume x, Number x, xxxx xxxx

Research Article

Enhancing Chinese Sarcasm Detection with A Large-Scale Manually Annotated Dataset and A Hybrid Deep Learning Model

Zixuan Wang, Zhiyong Feng*, Yan Zeng, Rafiul Haq, Xiping Zhao, Kaibing Yang, and Wenqi Cao

Abstract: Sarcasm detection is a critical aspect of sentiment analysis on social media platforms, where expressions often convey meanings that deviate from their literal interpretations. Although Chinese is considered a high-resource language, detecting sarcasm remains challenging due to the absence of large-scale, annotated datasets and sarcasm's complex, context-dependent, and culturally nuanced nature. This paper presents a multi-dimensional approach to sarcasm detection in Chinese, effectively addressing these challenges. We introduce Chinese TikTok Sarcasm Dataset (CTSD), a relatively large-scale, manually annotated collection of sarcastic comments sourced from TikTok, to address data scarcity. We demonstrate the effectiveness of the new dataset through extensive experiments using state-of-the-art models and introduce a hybrid deep learning model, BERTWWM-CNN-BiLSTM, which combines the power of Bidirectional Encoder Representations from Transformers with whole word masking (BERT-wwm) for rich contextual embeddings, Convolutional Neural Network (CNN) to capture local n-gram features, and Bidirectional Long Short-Term Memory network (BiLSTM) for learning long-range dependencies. Experimental results show that BERTWWM-CNN-BiLSTM outperforms baseline models, achieving an F1-score of 79.07% and an Area Under the ROC Curve (AUC) of 85.51%, reflecting improvements of 3.58% and 10.63%, respectively. These findings highlight the potential of our approach in enhancing sarcasm detection in social media text.

Key words: sarcasm detection; sentiment analysis; Chinese text; Natural Language Processing (NLP)

1 Introduction

Sarcasm detection is a fundamental challenge of Natural Language Processing (NLP) due to its reliance on implicit meanings, contextual cues, and linguistic subtleties that often contradict the literal interpretation of the text[1]. These challenges are amplified when applied to languages with deep cultural roots and complex linguistic features, such as Mandarin Chinese[2]. The Chinese writing system is especially challenging as certain characters are polysemous, the communication style is often indirect, and the meaning depends on the underlying intent. While sarcasm detection in English has seen significant advancements[3], current models struggle to effectively detect sarcasm in Chinese text. This gap highlights the need for novel approaches designed to address the distinctive linguistic characteristics of the Chinese language, ensuring more accurate identification of sarcasm.

• Zixuan Wang, Zhiyong Feng, and Rafiul Haq are with School of Computer Software, Tianjin University, Tianjin 300354, China. E-mail: a2414023626@tju.edu.cn; zyfeng@tju.edu.cn; Rafi@tju.edu.cn.
• Yan Zeng is with College of Life Sciences, Tianjin University, Tianjin 300354, China. E-mail: 3022208052@tju.edu.cn.
• Xiping Zhao and Wenqi Cao are with College of Mechanical Engineering, Tianjin University, Tianjin 300354, China. E-mail: 3021001669@tju.edu.cn; 3021001672@tju.edu.cn.
• Kaibing Yang is with College of Future Technology, Tianjin University, Tianjin 300354, China. E-mail: kaibing_yang@tju.edu.cn.
* To whom correspondence should be addressed.
Manuscript received: 2025-05-10; revised: 2025-08-16; accepted: 2025-09-12

Despite the growing interest in sarcasm detection, many challenges persist. A major issue is that sarcasm's cultural and contextual sensitivity makes it difficult for models to generalize across languages. Although Chinese is classified as a high-resource language, it presents specific difficulties in sarcasm detection. The inherent complexity of the language, combined with the scarcity of large-scale, sarcasm-annotated corpora, presents significant barriers for researchers seeking to develop effective models. Mandarin Chinese, the world's most spoken language[4], is a dominant medium for online discussions, further amplifying the challenges for NLP applications. Accurately detecting sarcasm in Chinese text is crucial for enhancing sentiment analysis, detecting misinformation, and understanding online communication.

The ability to detect sarcasm in Chinese has far-reaching implications across various fields. In brand management, accurately interpreting customer assertions, including sarcastic statements, is crucial for monitoring online reputation[5]. Social media analytics can gain deeper insights into public sentiment beyond simplistic categorizations of posts as "positive" or "negative"[6]. In political discourse, recognizing sarcasm helps provide a more nuanced understanding of public opinion on policies[7]. Additionally, detecting sarcasm greatly enhances human-computer interaction, enabling more responsive and natural interfaces[8]. The failure to accurately detect sarcasm leads to distortions in sentiment analysis, compromising the effectiveness of NLP applications across diverse domains.

Initially, sarcasm detection relied primarily on rigid systems built around predefined rules and manually selected text features[9, 10]. This approach was later replaced by machine learning techniques and, more recently, deep learning methods[11–13], marking a significant shift toward data-driven models. These advancements opened new possibilities for sarcasm detection in Chinese. Pre-trained language models[14, 15] have recently demonstrated significant performance on various NLP tasks. However, their effectiveness in detecting sarcasm in Chinese is still largely dependent on the quality and quantity of available training data. The scarcity of well-annotated Chinese datasets specifically for sarcasm detection remains one of the primary obstacles to progress in this area.

To address these challenges, we introduce the Chinese TikTok Sarcasm Dataset (CTSD), a large-scale, manually annotated corpus of sarcastic comments sourced from Douyin (the Chinese version of TikTok). Featuring 128 topics, CTSD provides a balanced distribution of sarcastic and non-sarcastic comments, ensuring meaningful diversity for robust model training and evaluation. To improve sarcasm detection in Chinese, we propose a hybrid deep learning model, named BERTWWM-CNN-BiLSTM, which combines various architectural advantages. The model integrates the context representation capabilities of the Bidirectional Encoder Representations from Transformers with whole word masking (BERT-wwm) pre-trained language model[16] with the effective feature extraction capabilities of Convolutional Neural Networks (CNNs) and the sequential modeling ability of Bidirectional Long Short-Term Memory networks (BiLSTM). BERT-wwm produces context-aware word embeddings that preserve fine-grained semantic information. The CNN module leverages salient local n-gram features related to sarcastic expressions, whereas the BiLSTM module is crucial to memory remote dependencies and sequential context within the text. Next, the output obtained from these components is concatenated and forwarded to the fully connected layers, which make the final classification.

We demonstrate through experimental testing on CTSD that the proposed model outperforms existing baselines, including BERT-wwm and other pre-trained models tailored to pre-screened datasets for sarcasm detection. BERTWWM-CNN-BiLSTM attains an F1-score of 79.07% and Area Under the ROC Curve (AUC) of 85.51%, exceeding the previous state-of-the-art method by 3.58% and 10.63%, respectively. These findings suggest that the proposed hybrid model outperforms previous deep learning or PLM-based approaches in capturing subtle sarcasm-related cues.

This paper presents the following key contributions:

- **Development of the CTSD dataset:** The CTSD is introduced, a large-scale, rigorously annotated corpus of sarcastic comments sourced from Douyin. CTSD exceeds prior Chinese datasets in size, features a balanced class distribution, and encompasses a broad range of current Chinese social media themes.
- **Introduction of an effective ensemble architecture for Chinese sarcasm detection:** We present BERTWWM-CNN-BiLSTM, an architecture that systematically combines BERT-wwm, CNN, and BiLSTM components to address the multi-faceted nature of sarcasm detection in Chinese text. The model

is integrated, which deliberately accounts for the linguistic characteristics of Chinese discourse and the challenging issues that arise when identifying sarcasm in online social media. The architecture demonstrates that a systematic combination of established components can achieve superior performance while providing a reliable validation framework for assessing dataset quality.

- **Comprehensive validation and analysis:** Experimental findings underscore the superior performance of the BERTWWM-CNN-BiLSTM architecture compared to established baselines. Achieving an F1-score of 79.07% and an AUC of 85.51%, the model significantly outperforms individual components and baseline Pre-trained Language Models (PLMs). These results provide empirical evidence for the effectiveness of the proposed approach in addressing the complexities of Chinese sarcasm detection.

The remainder of this paper is organized as follows. In Section 2, we review the relevant literature on sarcasm detection, focusing on the specific challenges and existing datasets for the Chinese language. In Section 3, we elucidate the proposed methodology. A detailed analysis of the experimental results and an evaluation of the proposed model's effectiveness are included in Section 4. Section 5 concludes the research and outlines potential avenues for future work.

2 Literature Review and Related Work

Sarcasm detection is a surprisingly nuanced and challenging task for NLP developers. This involves recognizing where communicative intent diverges considerably from the literal meaning that is communicated[17]. In English sarcasm detection, significant progress has been made[3], but with the Chinese language, it becomes immensely more challenging due to cultural context. Our review examines the development of Chinese sarcasm detection research over time and highlights the dataset constraints and methodology issues that further indicate the motivation for this paper.

2.1 Imperative of advanced chinese NLP and sarcasm detection

The global eminence of Mandarin Chinese, the language with the highest number of native speakers in the world[4], and its rapidly growing usage, becoming notable in the online sphere due to the broadening economic and cultural reach of its impact in China[18], underscores the urgent necessity for advanced NLP solutions. In this context, sentiment analysis plays a key role, and the more subtle task of sarcasm detection has become a particularly important area. Accurately detecting sarcasm is not just a technological challenge; it is essential for many applications to work properly. These applications find utility in public opinion forecasting[6], online reputation management[19], enhancing human-computer interaction paradigms[20], and political communication analysis[7]. Incorrectly interpreting sarcastic utterances can lead to significant distortions in sentiment analysis models, thereby undermining the usefulness and trustworthiness of NLP applications across various industries.

2.2 Datasets: Critical bottleneck in sarcasm detection

The performance of any NLP model, especially deep learning based ones, closely relies on high-quality and accurate annotated data[21]. Early research in sarcasm detection predominantly employed rule-based systems and manually engineered features, leveraging established lexical resources and discernible linguistic patterns[9, 10]. The advent of machine learning, followed by rapid advancements in deep learning[22, 23], has precipitated a transformative shift toward data-centric paradigms. This has been followed by the mass adoption of PLMs[14, 15], highlighting the importance of large, high-quality datasets to achieve good performance with proper fine-tuning. Yet, a continuing and significant barrier, especially in Chinese sarcasm detection, has been the scarcity of these important sources.

2.3 Intrinsic challenges in discernment of sarcasm within Chinese text

Identifying sarcasm correctly in Chinese text data is a series of obstacles, much more than in English. These hurdles stem from a unique combination of language specificities and entrenched cultural connotations.

Polysemy and implied meaning are ubiquitous in Chinese speakers, making it a unique challenge to extract sarcasm within speech. One of the most remarkable features of Chinese characters is their rich polysemy; a single character can convey multiple meanings, and the exact meaning cannot be separated from its context[24]. This inherent ambiguity is compounded by the fact that Chinese discourse is often predicated on implied meaning, making a literal interpretation inadequate and misleading in identifying sarcasm.

Another major challenge in Chinese text Sarcasm detection stems from the propagative influence of word segmentation uncertainty. Written Chinese has no explicit word delimiters, which introduces considerable complications. This property makes splitting words (an essential step in preprocessing most NLP tasks) highly error-prone and ambiguous[25]. These initial inaccuracies can propagate through subsequent analytical stages, thereby exerting a detrimental effect on the performance of downstream tasks, notably sarcasm detection.

Furthermore, the confluence of cultural specificity and indirect expressive norms presents unique challenges to detecting sarcasm in Chinese. Sarcasm in Chinese is often intricately interwoven with established cultural conventions, frequently employing idioms, allusions, and rhetorical devices that may remain opaque to individuals lacking intimate familiarity with the underlying cultural context, a category that includes most, if not all, current automated systems[26]. Moreover, a pervasive preference for indirect expression, particularly in articulating negative sentiments[27], further contributes to the subtlety of sarcastic utterances, making their accurate detection a significantly more formidable task.

Taken collectively, these linguistic and cultural intricacies underscore and amplify the critical need for high-quality, representative datasets that capture the broad typology of sarcasm encountered within Chinese.

2.4 Intrinsic limitations of existing Chinese sarcasm datasets

As established in Section 2.2, the efficacy of sarcasm detection models, particularly within the data-driven paradigm of contemporary NLP, is fundamentally contingent upon the quality and representational characteristics of the training data. Consequently, a rigorously constructed dataset constitutes not merely a supplementary resource, but rather the essential condition, upon which performant sarcasm detection models are predicated. However, the specific context of Chinese sarcasm detection presents unique challenges, primarily due to inherent limitations within existing datasets. Therefore, critically examining these limitations is essential to inform future research trajectories and the development of datasets.

The advancement of robust and generalizable Chinese sarcasm detection is constrained by intrinsic limitations within currently available datasets. While valuable contributions have been made to the field (Table 1), a critical analysis reveals pervasive shortcomings that impede the development of not only robust and generalizable, but also crucially, interpretable computational models. These limitations are multifaceted, encompassing considerations of dataset scale, class imbalance, the rigor of annotation methodologies, and domain-specific constraints.

To begin with, many datasets suffer from a scale problem: they are simply not large enough to train complex deep learning models well. Early and some recent manually annotated datasets, such as the one released by Liu et al.[28], which contains only 238 sarcastic instances, are too small to enable PLMs to learn the nuanced, context-dependent patterns of sarcastic expression. The lack of data not only exacerbates the risk of overfitting but also limits the model’s ability to generalize to new data it has never seen before.

Second, the significant class imbalance between sarcastic and non-sarcastic instances in many datasets has deleterious consequences, often leading to biased models. This bias, which is quite common in Liu et al.[28] and Lin and Hsieh[31], gives the model a bias

Table 1 Dataset statistics for Chinese sarcasm detection.

Dataset	Source	Number of sarcasm samples	Number of non-sarcasm samples	Total	Method
Liu et al.[28]	Sina Weibo	238	3621	3859	Manual
Liu et al.[28]	Tencent Weibo	359	5128	5487	Manual
Liu et al.[28]	Netease Forum	546	9810	10 356	Manual
Lu et al.[29]	Sina Weibo	2398	2398	4796	Manual
Gong et al.[30]	Guanchazhe	2486	2486	4972	Manual
Lin and Hsieh[31]	PTT	17 256	9373	26 629	Semi-Auto
Our balanced dataset	TikTok	8002	8002	16 004	Manual
Our whole dataset	TikTok	8002	20 628	28 630	Manual

towards the majority, i.e., the non-sarcastic class. This bias, inherent in its nature, significantly hinders the practical application of these models in real-world scenarios with diverse sarcasm distributions.

Furthermore, the understandable drive to create larger datasets has led to the use of semi-automatic annotation techniques in several instances. These methods often rely on readily available proxies, such as hashtags or emoticons, as surrogates for sarcastic intent. For instance, the Lin and Hsieh dataset[31] uses semi-automatic approaches. This gives a practical appeal but adds a non-trivial amount of noise and imprecision. These proxies are inadequate as reliable markers of true sarcasm, resulting in a significant amount of label contamination. This contamination leads to degraded performance and reliability of the models that have been trained.

Lastly, there is the ongoing issue of cross-domain generalizability, as models trained on data from one domain tend to struggle when applied to others, which is exacerbated by the context-dependent nature of sarcasm. Additionally, datasets constructed from specific domains often exhibit limited generalizability to other communicative contexts. The way in which sarcasm is linguistically and stylistically realized is heavily context-dependent, as it varies greatly with different topics, genres, and registers. Thus, a model learned on one domain, for example, product reviews, might not generalize well to another domain, such as political speech, which combines sarcasm in different ways and is governed by different social conventions. This underscores the urgent requirement for datasets that accurately reflect the diverse nature of online communication in several dimensions.

The inherent defects in current Chinese sarcasm datasets serve as a key bottleneck for developing this field. This necessitates a cooperative and heterogeneous approach to creating larger, more balanced, and more carefully annotated datasets representing the different forms of sarcasm present in diverse contexts, domains, and communicative styles.

2.5 Methodological limitations & architectural innovation

While the preceding section highlighted the critical limitations inherent in existing datasets, this section critically examines the prevailing methodologies used in Chinese sarcasm detection. Most modern methods are based on PLMs, particularly variants of Chinese-adapted BERT models[14], which are fine-tuned on sarcasm datasets. Even though a few studies involve additional elements or ensemble techniques[32, 33], significant methodological limitations remain. These limitations are further exacerbated by the intrinsic linguistic complexities of Chinese sarcasm and the previously delineated inadequacies in available data resources.

Existing PLMs face serious difficulty in understanding the contextual cues associated with sarcasm. While BERT provides semantically rich embeddings, it often fails to correctly capture the subtle and sometimes specific contextual cues needed for detecting sarcasm. Sarcasm relies on nuanced non-literal comprehension[34] and would therefore require a distinctive contextual analysis beyond the capabilities of generic PLMs.

In addition to that, the importance of local feature patterns is often underestimated by current models which further restricts accurate identification of sarcasm. Patterns of local n-grams, which are specific word combinations, and idioms, may represent hints of sarcasm, even if the words seem neutral in meaning. Traditional PLM methods often fail to leverage local signals effectively.

Existing models similarly struggle to capture the widely distributed dependencies across non-adjacent, semantically distant words or phrases that characterize sarcasm. PLMs focus more on nearby contexts[35] and do not capture long-distance correlations that are relevant to sarcasm detection. Additionally, methods that utilize PLMs fail to take into account sequential dynamics, the order in which words in a sentence appear (and the course of semantic meaning as more words are added) can contain critical context for detecting sarcasm.

These core methodological limitations, combined with the dataset deficiencies described earlier in this post, necessitate a paradigm shift to a more comprehensive and nuanced computational framework. This framework must address the lack of high-quality annotated data and develop more advanced feature extraction and contextual awareness mechanisms. The fundamental requirements for this type of approach are a good, high-quality representative data foundation and a synergetic fusion of complementary deep learning paradigms.

Fundamentally, we need a bedrock of high-quality, representative data. A larger, more diverse, and well-annotated dataset is essential for building robust, generalizable, and efficient sarcasm models.

This further highlights the need for a synergistic combination of complementary levels of learned representations provided by deep learning paradigms. This involves leveraging the advantages of PLMs at smartly capturing global contextual semantics and the features of other deep learning architectures. CNNs extract salient local features, and Recurrent Neural Networks, particularly BiLSTM, are adept at modeling sequential dependencies, capturing the temporal dynamics that characterize the discourse of sarcasm.

The BERTWWM-CNN-BiLSTM model presented in this work is a comprehensive and direct solution to these complex challenges. By synergistically integrating the representational power of a powerful PLM (BERT-wwm) with the feature extraction capabilities of CNN and BiLSTM components, the model simultaneously captures both global and local contextual information, effectively emulating crucial sequential relationships. This novel architectural design, in conjunction with the introduction of the new, large-scale, and carefully annotated CTSD, constitutes a substantial advancement in pursuing more accurate, robust, and interpretable Chinese sarcasm detection. The CTSD directly addresses the data set limitations outlined in Section 2.4, while the proposed model architecture decisively overcomes the methodological shortcomings identified in this section, paving the way for more reliable and effective sarcasm detection systems.

3 Methodology

The proposed approach is systematically structured into two key phases: (1) Data development, ensuring data quality and suitability for model training; (2) Model architectures development, focusing on extracting salient features using PLMs and employing a hybrid deep learning architecture to discriminate between sarcastic and non-sarcastic instances. These phases are detailed in the subsequent sections.

3.1 Dataset development

This research presents a sarcasm detection model trained on a curated Douyin comment corpus that represents contemporary Chinese online discourse. A bespoke Python web scraper facilitates data collection, systematically retrieving comments via Douyin's API from videos across 128 diverse topics, including gender equality, education, technology, and current affairs. Video prioritization within each topic favores those attracting the most comments, thereby enriching the dataset with varied linguistic patterns. Each comment's raw data include user ID, username, timestamp, IP location, like count, and the comment text. Preprocessing involves a multi-stage pipeline: newline removal, whitespace normalization, and emoji substitution with textual equivalents. URLs are retained for context, while other special characters are discarded.

Sarcasm annotation forms a critical component of this work. The establishment of the CTSD is facilitated by a highly structured, multi-step annotation workflow (Fig. 1), ensuring the quality and reliability of the data. Twelve native Chinese speakers with various professions undergo standardized training for detecting sarcasm as a first step. Four primary annotators, selected from this pool for their excellent performance in a rigorous preliminary evaluation, perform the main annotation task.

The annotation is developed based on a delineated two-step strategy to ensure consistency. As shown in Fig. 1, in Phase I, subsets of the comments are randomly sampled and assigned to a pair of annotators from the core team. These annotators work in isolation (blind annotation) to annotate each comment as being sarcastic or non-sarcastic. On each of these initial annotations, the Inter-Annotator Agreement (IAA) across pairs is computed using Cohen's Kappa (κ) statistic. A κ threshold of 0.6 is set as the cutoff for acceptable agreement, indicating moderate- to good-agreement. Any subset falling below this threshold is marked for discussion and potential re-annotation by both pairs to reconcile any disagreements. For $\kappa > 0.6$ subsets, comments with unary inter-annotator agreement are added directly to the final annotated dataset.

Comments in a sufficiently agreed upon subset ($\kappa > 0.6$), for which the initial two annotators return different labels, advanced to Phase II. In this step, the controversial comment is labeled by a third independent annotator from the core team. Finally, the description is formally annotated with the agreed-on label, according to the majority decision of the three annotators. This iterative, multi-stage process, which takes place over 32 days and involves over 380 person-hours of work, utilizes several expert viewpoints to ensure the high fidelity and consistency of the resulting

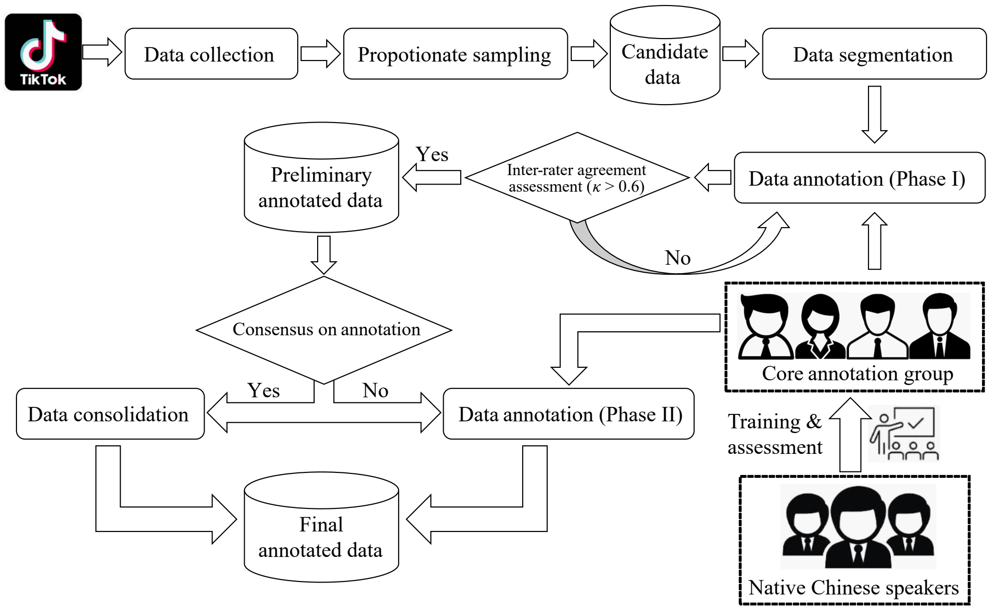

Fig. 1 Overall data annotation workflow.

annotations.

Of the nearly 30 000 comments, 8002 are deemed sarcastic. To minimize the impact of class imbalance, a balanced dataset of 16 004 comments (50% sarcastic and 50% non-sarcastic) is established by random under-sampling of the non-sarcastic majority. This stratified corpus is split into training (70%), validation (10%), and testing (20%) sets, while preserving the class proportionality. User identifiers and usernames are anonymized before model training and analysis.

3.2 Model architectures for sarcasm detection

The second phase is crucial to establishing a robust foundation for sarcasm detection in Chinese text. At this stage, we aim to leverage state-of-the-art PLMs to extract fine-grained linguistic features from comments and utilize a novel hybrid deep learning architecture for sarcasm detection. Through the use of contextualized embeddings incorporated into deep learning architectures, this approach accommodates the richness of sarcastic language which is characterized by the subtle use of semantic information and contextual features. The design is organized into two main sub-sections: feature extraction and representation learning, which discusses the selection and integration of PLMs, and sarcasm classification model, which describes the development and analysis of CNN, BiLSTM, and a hybrid of both. Such a holistic methodology enables us to model both local patterns and long-range contextual relations, allowing our model to capture sarcasm effectively in the volatile and heterogeneous Chinese cyberculture.

3.2.1 Feature extraction and representation learning

This research leverages the power of contextualized word embeddings, derived from PLMs, to address the task of sarcasm detection in Chinese text. This challenging task requires an understanding of subtle semantic nuances and contextual dependencies. State-of-the-art feature-based methods, such as Bag-of-Words, TF-IDF, and One-Hot Encoding, are insufficient because they fail to represent the syntactic richness and contextual dependencies inherent in sarcastic language. Built upon the large transformer architecture, BERT learns deep, bidirectional representations of text, which is a more effective method. These embeddings capture semantic similarity and context in relatively dense, low-dimensional vectors, which is a huge win compared to traditional approaches. This level of representational capacity is essential for understanding sarcasm, which is a nuanced form of communication that often relies on subtle linguistic signals and contextual knowledge.

Two paradigmatic usage strategies of PLMs, such as BERT, are fine-tuning and feature extraction. Fine-tuning means adding specific layers and training the full model, including the pre-trained weights. In contrast, feature extraction uses the PLM as a static feature encoder, providing embeddings without modifying the underlying pre-trained parameters. Our approach follows the latter, allowing for a direct evaluation of the representational power of different PLMs for the task of sarcasm detection.

An evaluation is conducted to determine the most suitable PLM for this purpose. We consider six widely used pretrained Chinese models: BERT, BERT-wwm, ELECTRA[36], XLNet[37], RoBERTa-wwm[38], and MacBERT[39]. Using the topic and corresponding post, each model creates embeddings from the input text, converting it into numerical representations of the text. The resulting embeddings are then used as input features for the downstream sarcasm classification model.

These six PLMs are selected based on their widespread use and demonstrate effectiveness in various Chinese NLP tasks. They represent diverse architectures and pre-training objectives, comprehensively evaluating the most advanced pre-trained Chinese models currently available.

The primary model employed in this investigation is the BERT-wwm model, a decision justified by its empirically superior performance relative to alternative PLMs. As delineated in Section 4.3, this model consistently yields higher F1-scores, a key performance indicator in this context. This statistically significant performance improvement suggests that the BERT-wwm model possesses architectural or pre-training characteristics that are particularly well-suited to detecting and classifying sarcastic utterances within the defined target domain.

3.2.2 Hybrid models

Our sarcasm detection framework encompasses three principal phases, depicted in Fig. 2.

(1) Contextualized embedding generation

The process begins with preprocessing raw text data, including the topic and its corresponding post. Tokenization is performed using the tokenizer corresponding to the BERT-wwm PLM. The resulting tokens are then input to the BERT-wwm architecture, a 12-layer transformer encoder network. We extract the sequence output from BERT-wwm, yielding a contextualized embedding for each input token. This output is structured as an $n \times d$ matrix, where n represents the token count and d represents the embedding dimensionality. At this level, each row of this matrix is a d-dimensional vector for a token's embedding. This sequence output, derived from BERT-wwm, provides the basis for the inputs of any subsequent deep learning model.

(2) Deep learning models

The next step involves employing three different deep learning models, each trained for feature extraction and classification on BERT embeddings. The first, BERTWWM-CNN, it uses a CNN-based model to extract local features from the BERT embeddings. The second, BERTWWM-BiLSTM, it applies a bidirectional LSTM Network on top of BERT embeddings for sequential relationships. The third, a hybrid model named BERTWWM-CNN-BiLSTM, it combines CNN and BiLSTM components, enriching

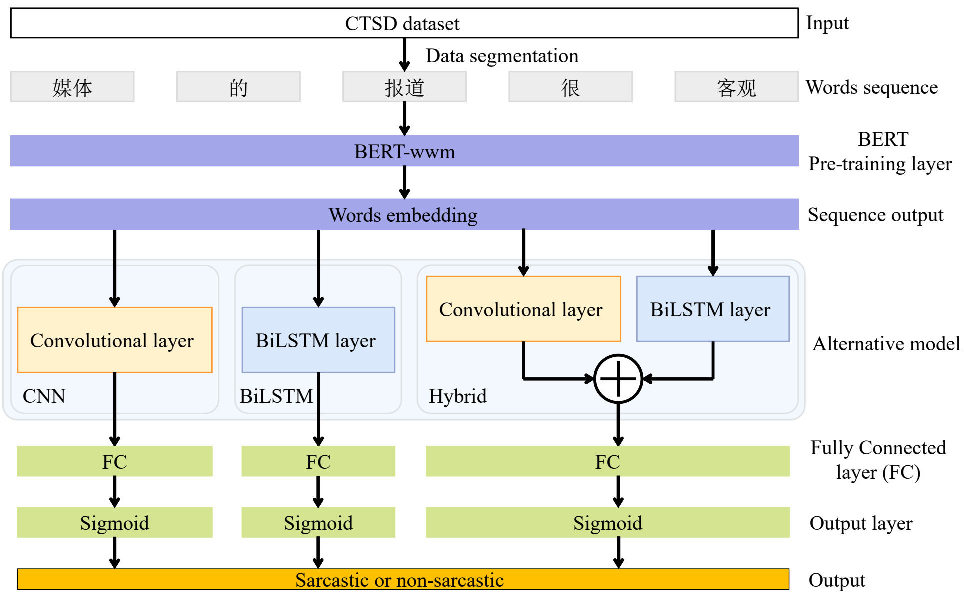

Fig. 2 Overall methodology diagram of the proposed architecture.

the BERT embeddings simultaneously and merging their corresponding outputs. Each model is trained to process the BERT embedding to help the downstream classification.

BERTWWM-CNN: Attempting to solve the inherently difficult problem of sarcasm detection effectively, we put forward the BERTWWM-CNN model curve, integrating the advantages of the deep semantic representation capacity of PLM BERTWWM and CNN's local feature extraction capacity. Sarcasm is a prime example of an abstract linguistic phenomenon that requires an understanding that cannot be deduced purely from semantics, as it approaches an overwhelming amount of contextual information that extends beyond semantic knowledge to convey implied meanings. BERTWWM generates context-aware word embeddings by leveraging large-scale pre-training, enabling it to predict subtle differences in the semantics of words across different contexts, which can significantly assist in understanding global semantics of sarcasm. At the same time, CNN is particularly adept at identifying local patterns in text, which is well-suited to detect strong signals of sarcasm, such as certain combinations of words or phrases.

This integrative strategy is reflected in the architecture of the BERTWWM-CNN model (shown in Fig. 3). The BERT encoding layer is designed to produce a global semantic representation for the input text. Expanding on that, the multi-scale convolutional layer starts to capture meaningful local characteristics. This convolutional layer incorporates three parallel kernels with varying receptive fields, capturing textual patterns within different length ranges. Subsequently, global max-pooling operations select the most representative features from each scale. This design accommodates long-range dependencies and local feature extraction, achieving a synergistic effect. Finally, a fully connected layer integrates all extracted features and outputs the probability of the input text belonging to the sarcasm class via a sigmoid activation function. Consequently, through the deep integration of BERTWWM and CNN, the BERTWWM-CNN model achieves an effective combination of semantic understanding and pattern recognition, leading to a more comprehensive and accurate characterization of Chinese text features and improved performance in sarcasm detection.

BERTWWM-BiLSTM: For binary sarcasm detection in Chinese text, a novel deep learning model named BERTWWM-BiLSTM is proposed (illustrated in Fig. 4). This model employs a hybrid architecture that integrates the advantages of PLMs and recurrent neural networks. Specifically, the model is based on the pre-trained BERT-wwm model, constructing contextually rich word vector representations to capture the subtle semantic nuances crucial for sarcasm detection. The resulting word vectors are then passed to a two-layer Bidirectional LSTM model capable of capturing long-range dependencies, endowing it with the greatest contextual awareness among the various architectures explored.

In the final stage of the model, several fully connected layers are implemented, along with a dropout layer for regularization. The fully connected

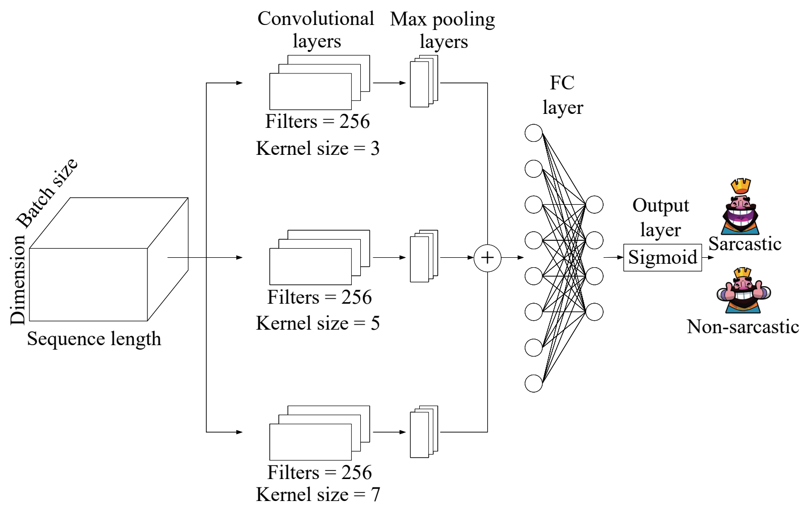

Fig. 3 Architecture of BERTWWM-CNN. Each convolutional block applies 256 filters over the token sequence, with kernel sizes of 3, 5, and 7.

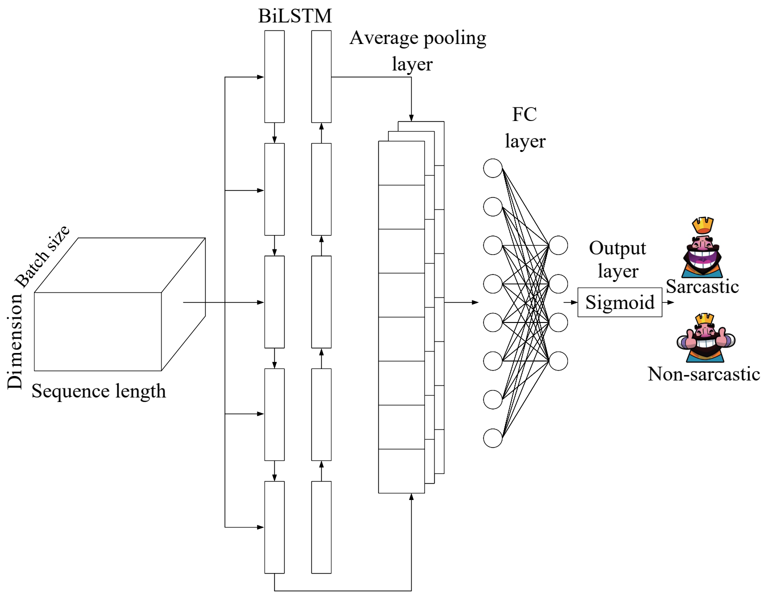

Fig. 4 Architecture of BERTWWM-BiLSTM.

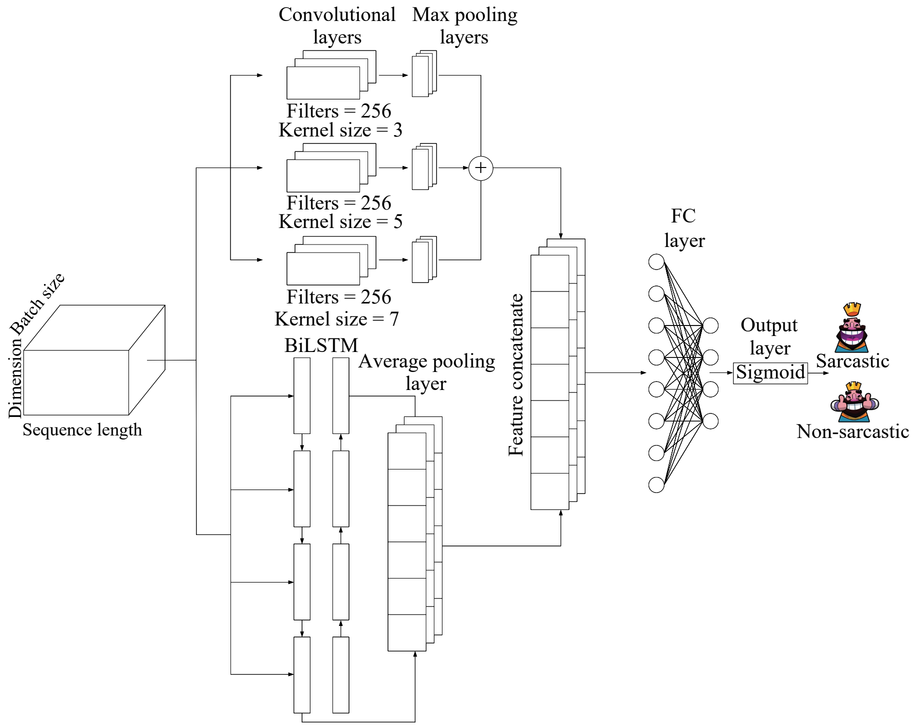

Fig. 5 Architecture of BERTWWM-CNN-BiLSTM. Each convolutional block applies 256 filters over the token sequence, with kernel sizes of 3, 5, and 7.

layers convert the output feature vectors generated by the BiLSTM network into a single scalar output, indicating the presence or absence of sarcasm. In BERTWWM-BiLSTM, the overall architecture is tailored specifically to tackle key challenges of sarcasm detection, such as sensitivity to contextual information, modeling of both local and long-range dependencies, and the potential class imbalance in the dataset. The thorough contextual understanding of BERT in conjunction with the sequential representation capability of BiLSTM enables the model to learn the rich patterns unique to sarcastic text.

Hybrid BERTWWM-CNN-BiLSTM model: We propose BERTWWM-CNN-BiLSTM (as depicted in

Fig. 4), a novel deep learning architecture for binary sarcasm detection in Chinese text, which fully considers the complexities and high dimensionality of this task. This model consists of a set of CNN and BiLSTM, which use the pre-trained BERT-wwm model to create contextualized word representations. Embedding helps in identifying intricate semantic details important for correctly identifying sarcasm.

The building blocks of architecture consist of two primary modules: a CNN-based module (BERTWWM-CNN) and a BiLSTM-based module (BERTWWM-BiLSTM). The BERTWWM-CNN module uses parallel convolutional layers with different kernel sizes to extract local n-gram features that may indicate sarcasm. The BERTWWM-BiLSTM module, on the other hand, is designed to capture long-range dependencies and sequential information to establish the broader textual context. These components operate independently, yielding distinct feature sets.

The final stage integrates these feature sets via concatenation, forming a comprehensive representation. This combined vector undergoes layer normalization and dropout for regularization, followed by processing through fully connected layers. The output is a single logit, signifying the presence or absence of sarcasm. By synergistically combining BERT's contextual understanding with the feature extraction capabilities of CNN and the sequential modeling of BiLSTM, BERTWWM-CNN-BiLSTM offers a robust and effective approach to Chinese sarcasm detection.

(3) Classification

In this final step, all three models employ an identical classification layer with sigmoid activation to output the predicted probabilities of sarcasm. This consistent approach ensures fair comparison between the different architectures (CNN, BiLSTM, and hybrid) while maintaining the same BERT embeddings as the input base.

4 Experiment and Evaluation

A rigorous comparison between the proposed model and a collection of well-established baselines is presented. All models are trained and assessed using the dataset detailed in Section 4.1, with a consistent 7:1:2 ratio for training, validation, and testing data. The preprocessing steps, outlined in Section 3.1, are applied uniformly to all input data. The experimental setup, including system specifications and a thorough investigation of the impact of hyperparameters on model effectiveness, is described in Section 4.2. Section 4.3 then details the empirical findings, providing an in-depth discussion and critical analysis of performance variations observed between the proposed methodology and the chosen baselines.

4.1 Datasets

This study utilizes the CTSD, presently the largest manually annotated collection of Chinese sarcastic comments. The data within CTSD originate from user-generated content on Douyin, TikTok's mainland Chinese equivalent. This dominant short-video platform in China boasts a substantial user base and exhibits exceptionally high engagement. A 2025 comparative analysis of short-video platforms[40] indicates that Douyin's monthly active users exceeded 800 million, and its daily active users surpassed 700 million by late 2024. The platform's user demographics are diverse, encompassing a wide range of age groups and geographical locations throughout China. These figures highlight its leading market position, showcasing a substantial competitive edge in user activity and reach. Therefore, the scale and engagement level of Douyin's users provide an invaluable dataset for analyzing current trends in Chinese online communication and socio-cultural interactions.

The CTSD encompasses 128 current topics spanning diverse fields, including entertainment, feminism, childbirth, society, culture, law, medicine, education, technology, finance, and diet, ensuring comprehensive representation of contemporary Chinese online discourse. The deliberate selection of topics for their controversial nature yields a high proportion of sarcasm within the collected comments.

Construction of the CTSD involved a multi-stage annotation procedure. Initially, 12 native Chinese speakers with diverse professional backgrounds were trained in sarcasm identification through a standardized protocol. Subsequently, a core group of four annotators, chosen for their superior performance on a rigorous evaluation, performed the final annotation (elaborated upon in Section 3.1). This careful selection process, spanning 32 days and over 380 person-hours, ensures the high fidelity of the annotations by integrating different perspectives and experiences.

Source data were derived from comments associated with the pre-selected Douyin trending topics. Of the

30 000 comments the researchers reviewed, 8002 were tagged as sarcastic. The equivalent number of non-sarcastic comments was randomly selected to prevent class imbalance and obtain a balanced dataset of 16 004 comments. This approach enables the establishment of solid and impartial sarcasm detection models. Therefore, the CTSD provides a valid and extensive resource for researching Chinese sarcasm detection on short-form social media text.

The dataset composition is illustrated in Table 2, providing insight into the distribution of each sub-topic and the total number of comments collected before balancing. This provides insight into the breadth of the dataset, including its coverage of various themes and the volume of user-generated content it encompasses.

Table 2 Distribution of topics and comments across different domains on CTSD.

Domain	Number of topics	Number of comments
Entertainment	10	2155
Feminism	5	1403
Fertility	4	708
Society	19	4036
Culture	13	2943
Law	12	2895
Medical	5	1331
Education	21	4793
Technology	14	2966
Finances	8	1714
Diet	9	2200
Emotion	3	421
Philosophy	5	1065
Total	128	28 630

The dataset comprises 128 distinct topics and 28 630 comments prior to balancing, as illustrated in Table 2. Such comment distribution across multiple domains reveals what content users are more likely to discuss. Some domains, such as "Education" and "Society", have a relatively large number of comments compared to others, like "Emotion" and "Fertility", where the input is smaller, but still valuable. Such variability in representation across various subjects is essential for properly capturing the nuances in online conversations and ensuring the construction of sarcasm detection models that can generalize across many different conversations. The balance approach described above produces a final dataset with 16 004 instances.

In addition, to better understand the dataset composition at the domain, we explore the distribution of comments labelled as sarcastic or non-sarcastic under each thematic category. The proportion of sarcastic comments in a representative sample of these domains is shown in Fig. 6.

There is substantial domain dependence in sarcasm prevalence, as observed in Fig. 6. The categories "Entertainment" and "Emotion" have a significantly higher proportion of sarcastic comments (35.3% and 35.4%, respectively) compared to the much lower rates in "Fertility" and "Medical" (24.4% and 22.5%, respectively). This pattern difference may suggest a correlation between the sensitivity of a topic and the tendency to express sarcasm. The "Entertainment" and "Emotion" domains are often narrowed to the subjective and personal, and they might elicit more sardonic replies, serving as a means of humor or

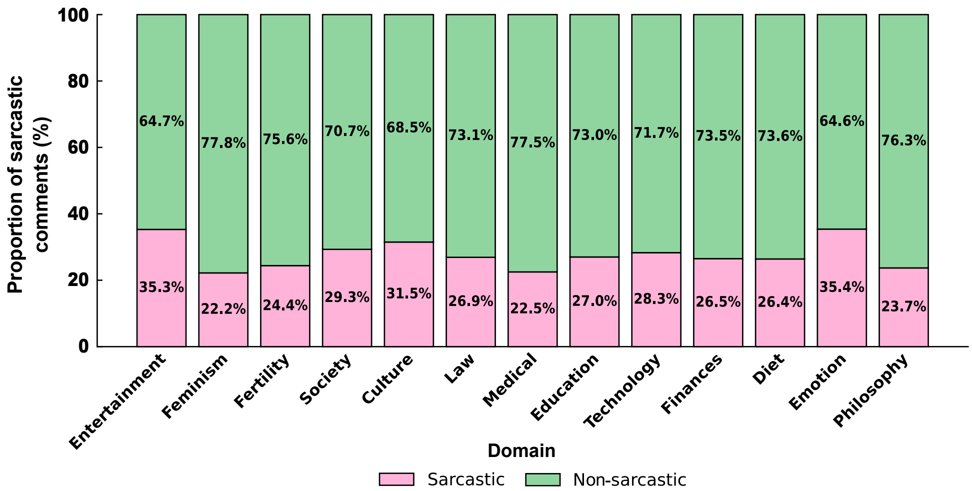

Fig. 6 Distribution of sarcastic comments across different domains on CTSD.

protest. On the contrary, the relatively well-established and solemn nature of the domains “Fertility” and “Medical” seems to correlate to diminished applications of sarcasm. The variability in sarcasm prevalence highlights the importance of considering domain-specific factors in sarcasm identification.

4.2 Experimental setup

In this study, we examine three BERT-based models, including a convolutional (BERT-CNN), a bidirectional LSTM (BERT-BiLSTM), and a hybrid model (BERT-CNN-BiLSTM) that synergistically utilizes the capabilities of the first two. All models are built and trained using the PyTorch framework, leveraging the computational capabilities of NVIDIA H800 GPUs. All the architectures have the BERT-wwm PLM as the foundational model. This specific BERT variant is fine-tuned for improved Chinese processing and implements whole-word masking, a method that has been proven to increase performance in downstream tasks drastically. We utilize the accompanying BERT Tokenizer, and the text is then transformed into a sequence normalized to 128 tokens, with padding or truncation applied as necessary. The dataset is split by 70% for training, 10% for validation, and 20% for testing, maintaining the same split across all experimental iterations to allow for valid comparisons.

We employ a multi-filter CNN to boost n-gram representation in the BERT-CNN model. They have three parallel convolutional layers with kernel sizes being 3, 5, and 7. After each convolutional layer, batch normalization and the max-pooling operations are applied, allowing the model to extract local contextual information in multiple granularities. This is followed by a fully connected layer and an output layer, both of which use a sigmoid function for probabilistic classification. The BERT-BiLSTM model, on the other hand, employs a BiLSTM network for modeling long-range, sequence-dependent relationships. This results in a sequence of hidden states from two layers of BiLSTM. These states are condensed using adaptive average pooling. Finally, similar to the BERT-CNN architecture, a fully connected layer with a ReLU activation and a subsequent sigmoid output layer performs the final classification task.

The hybrid model integrates features extracted from both the BERT-CNN and BERT-BiLSTM sub-models. Features from these sub-models are merged via concatenation before layer normalization and subsequently processed by dropout and fully connected layers. This design aims to create a more comprehensive representation, incorporating localized contextual information (from the CNN) and sequential patterns (from the BiLSTM).

All models are optimized using the AdamW optimizer, with an initial learning rate of 0.000 02. To ensure numerical stability during training, we employ binary cross-entropy loss through the BCEWithLogitsLoss function provided by the PyTorch library. We employ a combination of learning rate scheduling and early stopping to prevent overfitting and enhance generalization. All three models utilize the ReduceLROnPlateau scheduler, dynamically adjusting the learning rate based on validation set performance. Early stopping is implemented across all models, triggered by a lack of improvement in the validation F1-score over five epochs. A comprehensive overview of the hyperparameter settings for each model is presented in Table 3.

4.3 Results and discussion

To comprehensively assess the performance of the proposed models for Chinese sarcasm detection, a comparative analysis is presented in this section against a set of baseline methods from the existing literature. We evaluate our approach using the CTSD, a large-scale, manually annotated corpus specifically designed for this task (described in Section 3.1). Evaluation metrics for model performance are computed using a standard set of commonly used measures, selected based on their importance in the context of sarcasm detection: Precision (P), Recall (R), F1-score (F1), Area Under the ROC Curve (AUC), and Accuracy (Acc). Precision measures the ability to avoid false positives (labeling non-sarcastic text as sarcastic), and

Table 3 Hyperparameter setting.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	0.000 02
Batch size	16
Number of epochs	50
Dropout	0.3
Max sequence length	128
Loss function	Binary cross-entropy
Scheduler	ReduceLROnPlateau
Early stopping	Yes
Early stopping patience	5

Recall measures its ability to capture all true sarcastic text instances (minimizing false negatives). It is crucial for class imbalance cases, as the F1-score, the harmonic mean of Precision and Recall, helps provide a balanced measure of overall performance. The AUC is their rank of sarcastic instances above non-sarcastic instances, which gives its discriminative power. Finally, Accuracy provides the overall percentage of instances that are correctly classified. A complete presentation of the experimental results, including these metrics for all models, is provided in Table 4.

4.3.1 Baseline performance evaluation

To establish a rigorous baseline against which to compare the performance of our proposed models, we initially evaluate six widely used PLMs: BERT, BERT-wwm, ELECTRA, XLNet, RoBERTa-wwm, and MacBERT. These models, selected for their demonstrated effectiveness in various NLP applications[41], are subjected to a standardized fine-tuning procedure using the CTSD. Following fine-tuning, each PLM is directly applied to the sarcasm detection task on the CTSD test set. Among this collection of baseline models, BERT-wwm achieves the highest F1-score of 75.49%, providing a compelling justification for its selection as the core embedding layer in our subsequent model architectures. The other baseline PLMs exhibits F1-scores ranging from 72.47% to 74.5%. While these results demonstrate some proficiency in identifying sarcastic instances, they also highlight the considerable challenges associated with automated sarcasm detection, particularly in the context of social media platforms' short, informal text characteristics. This observation highlights the need for specialized model architectures that can capture the subtle linguistic and contextual cues often distinguishing sarcastic utterances from literal ones, a capability not fully provided by general-purpose pre-trained representations alone.

4.3.2 Performance evaluation of hybrid deep learning architectures

To address the performance limitations observed in the baseline PLM evaluation, we design and implement two distinct, task-specific deep learning architectures: BERTWWM-CNN and BERTWWM-BiLSTM. Both models utilize BERT-wwm embeddings but employ fundamentally different approaches to feature extraction and contextual representation, enabling a comparative analysis of their effectiveness in sarcasm detection.

BERTWWM-CNN: This architecture is predicated on employing CNN layers to extract salient local n-gram features. Evaluation of the CTSD reveales that BERTWWM-CNN achieves F1-score of 77.83%, Recall of 77.29%, and Precision of 78.38%. The balance of performance in the metrics demonstrates that the model is effective at both identifying true sarcastic cases and reducing false positives. The balance between precision and recall in these results indicates that the local linguistics pattern capacity of CNNs nicely complements the contextual representations produced by BERT-wwm. This balanced performance is visually confirmed by a confusion matrix (Fig. 7), which shows the proportionate distribution of errors (1706 false positives and 1817 false negatives), validating the robust classification ability of the model.

BERTWWM-BiLSTM: This model, in contrast, utilizes a BiLSTM network to capture long-range dependencies and sequential information within the

Table 4 Performance of different models on CTSD. Bolded F1-scores denote the best-performing baseline model and its three variants based on the model.

(%)

Model	Precision	Recall	F1-score	AUC	Accuracy
BERT	77.98	70.25	73.91	75.42	75.48
BERT-wwm	73.87	77.18	**75.49**	74.88	74.88
ELECTRA	74.59	74.41	74.50	74.57	74.57
XLNet	76.38	68.94	72.47	74.07	74.13
RoBERTa-wwm	76.37	70.86	73.51	73.95	73.88
MacBERT	75.66	70.39	72.93	74.20	74.26
BERTWWM-CNN	78.38	77.29	**77.83**	85.19	78.26
BERTWWM-BiLSTM	78.97	77.43	**78.19**	85.28	77.82
BERTWWM-CNN-BiLSTM	79.32	78.83	**79.07**	85.51	78.82

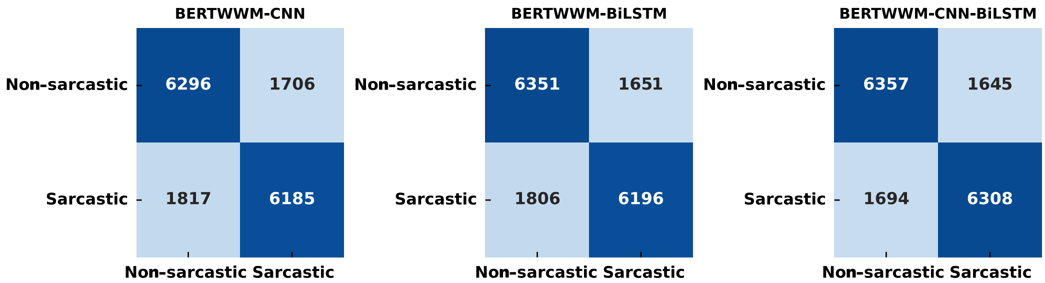

Fig. 7 Confusion matrices of proposed architectures on CTSD.

input text. BERTWWM-BiLSTM demonstrates stable performance, achieving F1-score of 78.19% and Precision of 78.97%. The improved precision score highlights the model's capability to reduce false positive classifications while maintaining high recall. This result provides strong evidence that modeling the broader context and sequential relationships between words is crucial for accurate sarcasm detection. We can see from Fig. 7 that the error distribution is very well balanced (1651 false positives and 1806 false negatives), indicating the improved discriminative capacity of the BiLSTM architecture in conjunction with BERT-wwm embeddings.

The substantial performance improvements achieved by both BERTWWM-CNN and BERTWWM-BiLSTM over all baseline PLMs underscore the critical importance of incorporating task-specific deep learning components that are meticulously tailored to the intricacies of sarcasm detection. These results not only highlight the complementary nature of local and global feature extraction, but also strongly suggest that a synergistic combination of these approaches, potentially within a unified architectural framework, may hold the key to achieving optimal performance.

4.3.3 Performance evaluation of the proposed BERTWWM-CNN-BiLSTM architecture

Building upon the insights gained from the individual BERTWWM-CNN and BERTWWM-BiLSTM models, we develop a hybrid architecture, BERTWWM-CNN-BiLSTM. The architecture receives the BERT-wwm embeddings as input, feeding them simultaneously into both the CNN and BiLSTM layers. The output feature vectors from the CNN and BiLSTM layers are then concatenated, followed by a fully connected layer with an activation function, yielding the final classification probability of sarcasm.

Among all the models, the hybrid BERTWWM-CNN-BiLSTM performs best in F1-score and AUC metrics. The best performing model, the hybrid model, achieves F1-score of 79.07% and AUC of 85.51%. Regarding the F1-score, it achieves a good balance of Precision (79.32%) and Recall (78.83%), outperforming the individual BERTWWM-CNN and BERTWWM-BiLSTM models, as well as all PLMs baselines. This balanced performance is further confirmed by the confusion matrix shown in Fig. 7, which contains only 1645 false positives and 1694 false negatives. The performance enhancement observed in sarcasm detection suggests that integrating local and global contextual information more effectively through the hybrid CNN-BiLSTM architecture is beneficial.

4.3.4 Comparative analysis of CTSD with available datasets

In comparison, two benchmark datasets are used to analyze the quality and robustness of our CTSD: ProsCons[42] and ToSarcasm[43]. We believe that this experiment serves two purposes: confirming the high quality of CTSD as a benchmark dataset and demonstrating that the performance of our proposed architecture is transferable and competitive across various sarcasm detection settings.

Table 5 presents a comprehensive comparison of performance metrics for the three datasets of the proposed model, utilizing the best architecture, BERTWWM-CNNBiLSTM. The empirical results show that CTSD enables significantly better model performance than both ProsCons and ToSarcasm on

Table 5 Performance comparison across datasets using BERTWWM-CNN-BiLSTM. The bold F1-score highlights the performance on CTSD, which is our main evaluation dataset.

(%)

Dataset	Precision	Recall	F1-score	AUC	Accuracy
ProsCons	70.71	51.37	59.51	87.16	85.57
ToSarcasm	76.82	77.62	77.22	72.14	73.10
CTSD	79.32	78.83	**79.07**	85.51	78.82

key evaluation metrics.

In particular, CTSD enables our model to achieve the highest F1-score (79.07%), which is 19.56 and 1.85% higher than ProsCons and ToSarcasm, respectively. This higher F1-score, obtained by a trade-off between high precision (79.32%) and reasonable recall (78.83%), suggests that CTSD represents a more balanced and representative corpus for sarcasm detection. The significantly higher precision score, in turn, reflects that CTSD annotations help the model learn better and more discriminative features, which result in fewer false positive classifications while maintaining stronger true positive detection rates.

Additionally, CTSD has AUC of 85.51%, which greatly improves upon ToSarcasm (72.14%) and is comparable to ProsCons (87.16%). This high AUC value indicates that CTSD helps the models to form strong discriminative features between sarcastic and non-sarcastic samples. The competitive accuracy of the dataset (78.82%) also indicates the high quality of the dataset and its usefulness in building a state-of-the-art sarcasm detection system. The overall improvements support the effectiveness of our careful data collection and annotation algorithms in Section 3.1 for CTSD, demonstrating that CTSD is a useful benchmark dataset for Chinese sarcasm detection research.

4.3.5 Cross-dataset state-of-the-art comparison

To comprehensively evaluate the effectiveness of CTSD as a benchmark dataset and assess the generalizability of our proposed architectures, we conduct extensive cross-dataset experiments. We evaluate four models—BERT-wwm, BERTWWM-CNN, BERTWWM-BiLSTM, and BERTWWM-CNN-BiLSTM—across three Chinese sarcasm detection datasets: ProsCons, ToSarcasm, and our proposed CTSD. This comprehensive evaluation, presented in Table 6, provides robust evidence for both the quality of CTSD and the effectiveness of our architectural innovations.

The results reveal several significant findings that validate CTSD as a superior benchmark for detecting Chinese sarcasm. First, all four tested models achieve their highest F1-scores when trained on CTSD compared to the other datasets. BERT-wwm achieves 75.49% on CTSD versus 56.60% on ProsCons and 74.33% on ToSarcasm. The performance gap is even more pronounced for our proposed architectures, with BERTWWM-CNN achieving 77.83% on CTSD compared to 57.21% and 75.62% on ProsCons and ToSarcasm, respectively. BERTWWM-BiLSTM exhibits similar patterns, achieving 78.19% on CTSD, compared to 58.22% and 76.12%. In contrast, our hybrid BERTWWM-CNN-BiLSTM reaches 79.07% on CTSD, outperforming 59.51% and 77.22% on the other datasets.

The cross-dataset evaluation also demonstrates the superiority of our proposed architectures over previous state-of-the-art models. On the ProsCons dataset, where MacBERT previously achieves the best performance with F1-score of 57.90%, all our tested

Table 6 Comprehensive performance comparison across datasets and models. Bold F1-scores denote the best-performing model on each dataset.

(%)

Dataset	Model	Precision	Recall	F1	AUC	Accuracy
ProsCons	MacBERT (previous SOTA)	69.50	49.80	57.90	–	–
	BERT-wwm	68.70	48.10	56.60	–	–
	BERTWWM-CNN	67.12	49.85	57.21	86.47	85.23
	BERTWWM-BiLSTM	68.46	50.64	58.22	85.04	85.22
	BERTWWM-CNN-BiLSTM	70.71	51.37	**59.51**	87.16	85.57
ToSarcasm	TOSPrompt (previous SOTA)	70.02	76.68	73.20	–	71.76
	BERT-wwm	76.38	72.38	74.33	70.27	70.64
	BERTWWM-CNN	75.42	75.81	75.62	69.22	70.56
	BERTWWM-BiLSTM	75.34	76.92	76.12	70.55	71.66
	BERTWWM-CNN-BiLSTM	76.82	77.62	**77.22**	72.14	73.10
CTSD	BERT-wwm	73.87	77.18	75.49	74.88	74.88
	BERTWWM-CNN	78.38	77.29	77.83	85.19	78.26
	BERTWWM-BiLSTM	78.97	77.43	78.19	85.28	77.82
	BERTWWM-CNN-BiLSTM	79.32	78.83	**79.07**	85.51	78.82

models show improvements. Even the baseline BERT-wwm model achieves comparable performance at 56.60%, while our architectural enhancements further improve performance, with BERTWWM-CNN-BiLSTM reaching 59.51%. This represents a 1.61% improvement over the previous state-of-the-art.

On the ToSarcasm dataset, where TOSPrompt establishes the previous benchmark at 73.20% F1-score, our results are even more impressive. All four models surpass this benchmark, with BERT-wwm achieving 74.33% and our hybrid BERTWWM-CNN-BiLSTM reaching 77.22%, a substantial 4.02% improvement. These consistent improvements across different datasets underscore the robustness and generalizability of our proposed architectures. It is worth noting that ToSarcasm's relatively smaller scale may contribute to its overall lower performance levels compared to CTSD; yet, our models still demonstrate clear advantages over the previous state-of-the-art.

Several reasons can explain the better performance of models trained on CTSD. Since CTSD is on a larger scale, it has more diverse training examples, giving better distributions of sarcastic patterns for the models to learn. The balanced class distribution also keeps models from biasing too far in favor of the majority class, resulting in more fair trade-offs between precision and recall. Moreover, the CTSD labels have been obtained through a thorough manual annotation process, which involves multiple annotators and quality checks, to provide high-quality labels that yield better training signals.

An analysis of the precision-recall trade-offs across datasets reveals that models trained on CTSD achieve more balanced performance. For instance, BERTWWM-CNN-BiLSTM trained on CTSD achieves Precision of 79.32% and Recall of 78.83%, demonstrating an excellent balance. In contrast, when trained on ProsCons, the same model shows a significant gap between Precision (70.71%) and Recall (51.37%), indicating difficulty in achieving balanced detection. This pattern holds across all tested architectures, suggesting that CTSD's characteristics enable more robust and balanced model training.

The AUC scores further support the effectiveness of CTSD as a training dataset. Models trained on CTSD consistently achieve higher AUC scores, with BERTWWM-CNN-BiLSTM reaching 85.51% compared to 87.16% on ProsCons and 72.14% on ToSarcasm. While ProsCons shows a slightly higher AUC, this is offset by its significantly lower F1-score, suggesting that the high AUC may be influenced by the dataset's severe class imbalance rather than genuine discriminative capability.

Our overall findings provide strong empirical evidence supporting the proposal that CTSD is indeed a good candidate for serving as a benchmark dataset of Chinese sarcasm detection research. These gains in performance, achieved with diverse architectures ranging from individual pre-trained models to advanced hybrid systems, provide consistent improvements, suggesting that the design choices of CTSD lead to more efficient model training. Therefore, CTSD stands to be highly advantageous for research on Chinese sarcasm detection, allowing methodological approaches of different researchers to be compared equitably.

4.3.6 Model performance analysis and practical implications

The comparative performance of the proposed models (BERTWWM-CNN, BERTWWM-BiLSTM, and the hybrid BERTWWM-CNN-BiLSTM) against the baseline PLMs highlights the importance of a specific type of deep learning architecture in sarcasm detection. PLMs serve as a powerful base of contextualized embeddings, but performance is significantly improved through additional layers that extract features that align well with the subtle linguistic signals of sarcasm. An exemplary application of this principle is presented in the hybrid model, which integrates local (n-gram) and global (long-range dependency) feature extraction in a synergic way. This suggests that effective sarcasm detection necessitates a comprehensive approach, integrating contextual knowledge from both local and global perspectives.

These outcomes support the proposed gathering approach for Chinese sarcasm detection on CTSD. This model uses BERTWWM with CNN and BiLSTM layers to get more powerful sarcasm features. However, these results are context-specific to the CTSD. The models must be further validated on more Chinese sarcasm datasets to confirm their generalizability. Additionally, the dynamic nature of online discourse necessitates ongoing monitoring and adaptation of models to ensure optimal performance in practice.

The practical significance of this research extends to a wide range of NLP applications. Accurate sarcasm detection is crucial for:

Enhanced sentiment analysis: Misinterpreting sarcasm can lead to significant errors in sentiment analysis, impacting the Accuracy of downstream tasks like opinion mining and market research. Our model improves the reliability of sentiment analysis in Chinese text.

Enhanced human-computer interaction: The nuanced understanding of sarcasm enables computational systems to engage in more natural and intuitive interactions, resulting in significant improvements in user experience for applications such as conversational agents (chatbots) and virtual assistants.

Online reputation management: Identifying sarcastic comments directed at brands or individuals allows for more effective monitoring and management of online reputation.

Granular social media intelligence: A fine-grained analysis of sarcasm's prevalence and contextual targets within social media corpora provides invaluable insights into the evolving landscape of public opinion, cultural trends, and socio-political dynamics.

4.3.7 Case study: Model performance on Chinese-specific sarcasm

To illustrate how our hybrid architecture addresses the unique challenges of Chinese sarcasm detection, we present a comparative analysis using representative sarcastic examples from CTSD. These examples, drawn from actual user comments on Douyin, demonstrate the distinct contributions of each architectural component and their synergistic effect when combined. Table 7 presents the sarcasm detection confidence scores generated by each model for these examples. These confidence scores represent the model's sarcasm probability output on a 0−1 scale, where values above 0.5 indicate sarcastic classification and higher values reflect stronger classification confidence.

Consider the first example from a discussion about the meaning of work: "My experience has enriched considerably, and every day at work refreshes my cognitive lower limits". This comment exemplifies the challenge of detecting sarcasm through surface-level positive sentiment that masks criticism. The phrase "enriched experience" appears positive, yet the subsequent revelation about "refreshing cognitive lower limits" creates a stark contradiction that signals sarcastic intent.

The baseline BERT-wwm model struggles with this example, generating a confidence score of 0.482, which falls below the 0.5 threshold and thus misclassifying it as non-sarcastic. This failure illustrates the model's susceptibility to surface-level positive lexical items, being misled by the initial positive framing. In contrast, BERTWWM-CNN shows significant improvement with a confidence of 0.627, correctly identifying the sarcastic intent by capturing the local pattern of "refreshing … lower limit", a phrase structure commonly used for sarcastic expression in Chinese internet discourse. The improvement of 0.145 in confidence demonstrates the CNN's effectiveness in recognizing local semantic contradictions that the baseline model missed. BERTWWM-BiLSTM achieves a slightly higher confidence of 0.641 by modeling the semantic progression from apparent enrichment to actual degradation, showcasing the BiLSTM's capability to track the expectation-reversal narrative across the sentence. The hybrid BERTWWM-CNN-BiLSTM achieves optimal performance with a confidence of 0.712, demonstrating a substantial improvement of 0.230 over the baseline by simultaneously leveraging both local pattern recognition and global semantic understanding.

A second example demonstrates sarcasm through absurd analogy: "Isn't this just "to prevent you from being robbed, give me your money first"?". This comment about rhino horn removal uses reductio ad absurdum to expose flawed logic. The sarcasm

Table 7 Model confidence scores on representative Chinese sarcastic examples.

Chinese sarcastic examples from CTSD	BERT-wwm	BERTWWM-CNN	BERTWWM-BiLSTM	BERTWWM-CNN-BiLSTM
My experience has enriched considerably, and every day at work refreshes my cognitive lower limits.	0.482	0.627	0.641	0.712
Isn't this just "to prevent you from being robbed, give me your money first"?	0.543	0.681	0.694	0.758
Didn't the heating company say: heating isn't to make everyone warm, but to keep everyone from being cold?	0.496	0.653	0.668	0.734

operates by creating a parallel between wildlife protection measures and robbery, revealing the absurdity of "protective" actions that cause the very harm they claim to prevent.

For this example, the baseline BERT-wwm achieves a marginal confidence of 0.543, barely exceeding the classification threshold. While correctly classified as sarcastic, the low confidence indicates uncertainty in the model's decision, likely due to partial recognition of the rhetorical question structure and quotation marks without fully comprehending the logical absurdity. The CNN component in BERTWWM-CNN achieves a more decisive confidence of 0.681, representing a 0.138 improvement through effective identification of multiple sarcastic markers: the rhetorical frame "isn't this just", the quotation marks signaling irony, and the parallel structure between "prevent … being robbed" and "give me money first". The BiLSTM component in BERTWWM-BiLSTM captures the logical progression with a confidence of 0.694, demonstrating its ability to understand how the preventive measure paradoxically mirrors the harm it claims to prevent. The hybrid model achieves the highest confidence of 0.758 by combining pattern recognition (rhetorical questions and quotations) with logical comprehension of the analogical reasoning that creates the sarcastic effect.

A third example showcases bureaucratic doublespeak sarcasm: "Didn't the heating company say: heating isn't to make everyone warm, but to keep everyone from being cold?". This comment uses verbatim quotation of the official rationalization to highlight service failures. The sarcasm arises from presenting the bureaucratic distinction between "warm" and "not cold" as a legitimate differentiation rather than a semantic manipulation.

The BERT-wwm model fails to detect the sarcasm with a confidence of 0.496, misclassifying this example as non-sarcastic. This failure highlights the model's inability to recognize when official discourse becomes self-parodying through semantic manipulation, interpreting the bureaucratic distinction as potentially legitimate rather than absurd. The CNN component achieves a confidence of 0.653, a significant 0.157 improvement, by identifying the quotation structure "didn't they say" and the reframing pattern "not … but …", both of which serve as strong indicators of ironic citation in Chinese online discourse. The BiLSTM component demonstrates a confidence of 0.668, successfully capturing the semantic equivalence between "warm" and "not cold" and understanding how the bureaucratic language attempts to redefine service standards downward. The hybrid architecture achieves a confidence of 0.734, combining structural pattern recognition with semantic comprehension to achieve a 0.238 improvement over the baseline, effectively identifying how official discourse becomes self-satirizing through its linguistic contortions.

These case studies reveal critical insights about our architectural design. The CNN component demonstrates particular effectiveness in capturing local sarcastic markers specific to Chinese online discourse, including quotation patterns, rhetorical structures, and characteristic phrase combinations. The BiLSTM component proves essential for understanding the narrative structures and logical relationships that transform surface statements into sarcasm, including expectation-reversal arcs, analogical reasoning, and semantic manipulation. Most importantly, the hybrid architecture achieves superior performance through the synergistic combination of these capabilities, enabling the robust identification of sophisticated sarcastic strategies, ranging from subtle workplace criticism to logical absurdity and bureaucratic self-parody.

The success of our hybrid approach in these various cases validates our design philosophy, which is based on Chinese language features. Each component contributes distinct capabilities, providing comprehensive coverage of the multifaceted nature of Chinese sarcasm. This targeted design approach, grounded in linguistic analysis and validated through empirical results, represents our primary methodological contribution in addressing the unique challenges of Chinese sarcasm detection.

4.3.8 Limitations and future research directions

While this research makes substantial progress, several limitations and avenues for future work remain:

Dataset generalizability and extensibility: Although the CTSD is large and diverse, its focus on Douyin comments may limit its direct applicability to other text genres or platforms. Future research must rigorously evaluate model performance across a comprehensive Chinese corpus, encompassing diverse social media platforms, literary works, and transcribed spoken language. Crucially, developing robust cross-domain adaptation techniques is essential for achieving generalizable models that can handle the inherent variability across disparate textual styles and domains.

Multimodal sarcasm detection: While this study focuses exclusively on textual sarcasm detection, online discourse is inherently multimodal. Sarcasm is often conveyed not through text alone but through the complex interplay of textual, visual, and auditory cues. Future research should investigate multimodal approaches that integrate these additional modalities to achieve more robust and accurate detection, particularly in platforms like Douyin.

Fine-grained sarcasm analysis: While binary classification (sarcastic vs. non-sarcastic) provides a foundational understanding, a comprehensive treatment of sarcasm demands a more granular approach. Beyond binary classification (sarcastic/non-sarcastic), future work could explore finer-grained analyses of sarcasm, such as identifying the target of the sarcasm, classifying the type of sarcasm (e.g., verbal irony, rhetorical questions, understatement), and quantifying the intensity or degree of sarcasm.

Adaptive modeling for dynamic online discourse: The relentless evolution of online language, marked by the ceaseless influx of neologisms, idiomatic expressions, and fluctuating communication patterns, presents a formidable obstacle to maintaining robust model performance. Future research should prioritize the development of adaptive modeling techniques that can be continuously refined. This necessitates exploring online learning paradigms and integrating external knowledge resources, such as curated linguistic corpora and real-time social media data streams. Such integration will empower models to dynamically assimilate emerging linguistic trends, ensuring sustained Accuracy and resilience in the face of linguistic drift.

5 Conclusion and Future Work

In this paper, we present the CTSD, the largest manually annotated dataset for detecting sarcasm in Chinese. CTSD corrects critical gaps in existing datasets and offers balanced classes, more topical diversity, and greater scale. This dataset provides a solid foundation for building and evaluating sarcasm detection systems with superior generalization capabilities.

Our proposed BERTWWM-CNN-BiLSTM architecture represents a significant methodological advancement, synergistically combining three complementary components: BERT-wwm for contextual embeddings, CNN for local n-gram feature extraction, and BiLSTM for capturing long-range dependencies. This ensemble approach effectively models the multi-faceted linguistic cues characteristic of sarcastic expressions in Chinese.

Experimental evaluation shows the efficacy of our approach, where the BERTWWM-CNN-BiLSTM model reaches F1-score and AUC of 79.07% and 85.51%, respectively, outperforming all baselines. Our model outperforms the best baseline (BERT-wwm) by 3.58% in F1-score and 10.63% in AUC, indicating that it generalizes well in identifying subtle sarcastic markers in the Chinese language.

Future directions for research will involve (1) studying cross-domain generalization for models able to transfer across different discursive contexts; (2) using multimodal analyses that incorporate visual and audio signals that convey non-verbal signs of sarcasm; (3) analyzing linguistic drift in sarcastic expression to implement models that adapt to linguistic evolution; and (4) extending the architecture to other languages with comparable levels of pragmatic complexity. These improvements would also make sarcastic detection systems more useful across various applications, such as sentiment analysis, web analytics, social media intelligence, and human-machine interaction.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (NSFC) (No. 62372323).

References

[1] C. F. Burgers and G. J. Steen, Introducing a three-dimensional model of verbal irony: Irony in language, in thought, and in communication, in *Irony in Language Use and Communication*, A. Athanasiadou and H. L. Colston, eds. Philadelphia, PA, USA: John Benjamins Publishing Company, 2017, pp. 87–108.

[2] A. Reyes and P. Rosso, On the difficulty of automatically detecting irony: Beyond a simple case of negation, *Knowledge Inf. Syst.*, vol. 40, no. 3, pp. 595–614, 2014.

[3] H. Yaghoobian, H. R. Arabnia, and K. Rasheed, Sarcasm detection: A comparative study, arXiv preprint arXiv: 2107.02276, 2021.

[4] Y. C. Wang, C. K. Wu, and R. T. H. Tsai, Cross-language article linking with different knowledge bases using bilingual topic model and translation features, *Knowledge-Based Syst.*, vol. 111, pp. 228–236, 2016.

[5] B. Liu and L. Zhang, A survey of opinion mining and sentiment analysis, in *Mining Text Data*, C. C. Aggarwal and C. X. Zhai, eds. New York, NY, USA: Springer, 2012, pp. 415–463.

[6] B. Liu, S. Tang, X. Sun, Q. Chen, J. Cao, J. Luo, and S. Zhao, Context-aware social media user sentiment analysis, *Tsinghua Science and Technology*, vol. 25, no. 4, pp. 528–541, 2020.

[7] D. Davidov, O. Tsur, and A. Rappoport, Enhanced sentiment learning using twitter hashtags and smileys, in *Proc. Coling 2010: Posters*, Beijing, China, 2010, pp. 241–249.

[8] C. I. Eke, A. A. Norman, L. Shuib, and H. F. Nweke, Sarcasm identification in textual data: Systematic review, research challenges and open directions, *Artif. Intell. Rev.*, vol. 53, no. 6, pp. 4215–4258, 2020.

[9] E. Riloff, A. Qadir, P. Surve, L. De Silva, N. Gilbert, and R. Huang, Sarcasm as contrast between a positive sentiment and negative situation, in *Proc. 2013 Conf. Empirical Methods in Natural Language Processing*, Seattle, WA, USA, 2013, pp. 704–714.

[10] C. I. Eke, A. A. Norman, and L. Shuib, Context-based feature technique for sarcasm identification in benchmark datasets using deep learning and BERT model, *IEEE Access*, vol. 9, pp. 48501–48518, 2021.

[11] A. Joshi, P. Bhattacharyya, and M. J. Carman, Automatic sarcasm detection: A survey, *ACM Comput. Surveys*, vol. 50, no. 5, p. 73, 2017.

[12] M. Pichler and F. Hartig, Machine learning and deep learning—A review for ecologists, *Methods Ecol. Evol.*, vol. 14, no. 4, pp. 994–1016, 2023.

[13] R. Misra and P. Arora, Sarcasm detection using hybrid neural network, arXiv preprint arXiv: 1908.07414, 2022.

[14] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, in *Proc. 2019 Conf. North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1* (*Long and Short Papers*), Minneapolis, MN, USA, 2019, pp. 4171–4186.

[15] Y. Bie, Y. Yang, and Y. Zhang, Fusing syntactic structure information and lexical semantic information for end-to-end aspect-based sentiment analysis, *Tsinghua Science and Technology*, vol. 28, no. 2, pp. 230–243, 2023.

[16] Y. Cui, W. Che, T. Liu, B. Qin, and Z. Yang, Pre-training with whole word masking for Chinese BERT, *IEEE/ACM Trans. Audio Speech Lang. Process.*, vol. 29, pp. 3504–3514, 2021.

[17] L. Zou, Creative computing in language understanding: A novel approach to pragmatic analysis, Ph.D. dissertation, University of Leicester, Leicester, UK, 2024.

[18] D. Guthrie, *China and Globalization: The Social, Economic and Political Transformation of Chinese Society*. 3rd ed. New York, NY, USA: Routledge, 2012.

[19] A. Zubiaga, A. Aker, K. Bontcheva, M. Liakata, and R. Procter, Detection and resolution of rumours in social media: A survey, *ACM Comput. Surveys*, vol. 51, no. 2, p. 32, 2018.

[20] H. F. T. Alsaadawi, B. Das, and R. Das, TAC-Trimodal affective computing: Principles, integration process, affective detection, challenges, and solutions, *Displays*, vol. 83, p. 102731, 2024.

[21] Y. Huang, K. Tang, M. Chen, and B. Wang, A comprehensive survey on evaluating large language model applications in the medical industry, arXiv preprint arXiv: 2404.15777, 2024.

[22] A. Ghosh and T. Veale, Fracking sarcasm using neural network, in *Proc. 7th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, San Diego, CA, USA, 2016, pp. 161–169.

[23] C. Peng, C. Zhang, X. Xue, J. Gao, H. Liang, and Z. Niu, Cross-modal complementary network with hierarchical fusion for multimodal sentiment classification, *Tsinghua Science and Technology*, vol. 27, no. 4, pp. 664–679, 2022.

[24] H. Chen, X. Xu, H. Li, X. Yu, R. Pan, and Z. Zhang, A database of ambiguous Chinese characters with measures for meaning dominance and meaning balance, *Appl. Psycholinguist.*, vol. 45, no. 4, pp. 695–716, 2024.

[25] P. C. Chang, M. Galley, and C. D. Manning, Optimizing Chinese word segmentation for machine translation performance, in *Proc. 3rd Workshop on Statistical Machine Translation*, Columbus, OH, USA, 2008, pp. 224–232.

[26] Y. Cao, Humour and (im)politeness in Chinese sitcom discourse: A case study of Ipartment, Ph.D. dissertation, Western Sydney University, Sydney, 2019.

[27] N. M. Lou and K. A. Noels, Sensitivity to language-based rejection in intercultural communication: The role of language mindsets and implications for migrants' cross-cultural adaptation, *Appl. Linguist.*, vol. 40, no. 3, pp. 478–505, 2019.

[28] P. Liu, W. Chen, G. Ou, T. Wang, D. Yang, and K. Lei, Sarcasm detection in social media based on imbalanced classification, in *Proc. 15th Int. Conf. Web-Age Information Management*, Macau, China, 2014, pp. 459–471.

[29] X. Lu, Y. Li, and S. Wang, Linguistic features enhanced convolutional neural networks for irony recognition, (in Chinese), *J. Chin. Inf. Process.*, vol. 33, no. 5, pp. 31–38, 2019.

[30] X. Gong, Q. Zhao, J. Zhang, R. Mao, and R. Xu, The design and construction of a Chinese sarcasm dataset, in *Proc. 12th Language Resources and Evaluation Conf.*, Marseille, France, 2020, pp. 5034–5039.

[31] S. K. Lin and S. K. Hsieh, Sarcasm detection in Chinese using a crowdsourced corpus, in *Proc. 28th Conf. Computational Linguistics and Speech Processing*, Tainan, China, 2016, pp. 299–310.

[32] X. Zeng, Few-shot claim verification for automated fact checking, Ph.D. dissertation, Queen Mary University of London, London, 2024.

[33] V. Fortineau, T. Paviot, and S. Lamouri, Automated business rules and requirements to enrich product-centric information, *Comput. Ind.*, vol. 104, pp. 22–33, 2019.

[34] T. Bali, Understanding pretense: A study in humour and

sarcasm, Ph.D. dissertation, International Institute of Information Technology, Hyderabad, India, 2021.
[35] Z. Yang, Z. Dai, Y. Yang, J. Carbonell, R. Salakhutdinov, and Q. V. Le, XLNet: Generalized autoregressive pretraining for language understanding, in *Proc. 33^{rd} Int. Conf. Neural Information Processing Systems*, Vancouver, BC, Canada, 2019, pp. 5753–5763.
[36] K. Clark, M. T. Luong, Q. V. Le, C. D. Manning, ELECTRA: Pre-training text encoders as discriminators rather than generators, arXiv preprint arXiv:2003.10555, 2020.
[37] Z. Yang, Z. Dai, Y. Yang, J. Carbonell, R. Salakhutdinov, Q. V. Le, XLNet: Generalized autoregressive pretraining for language understanding, in *Proc. Advances in Neural Information Processing Systems*, arXiv preprint arXiv:1906.08237, 2019.
[38] Y. Cui, W. Che, T. Liu, B. Qin, and Z. Yang, Pre-training with whole word masking for chinese BERT, *IEEE/ACM Trans. Audio Speech Lang. Process.*, vol. 29, pp. 3504-3514, 2021.
[39] Y. Cui, W. Che, T. Liu, B. Qin, S. Wang, and G. Hu, Revisiting pre-trained models for Chinese natural language processing, arXiv preprint arXiv:2004.13922, 2020.
[40] S. Wang, “When pigs fly”: Resources swapping, affordable marketing, and the transformation of Douyin from short video sharing to online shopping, *New Med. Soc*., doi: 10.1177/14614448251314448.
[41] X. Qiu, T. Sun, Y. Xu, Y. Shao, N. Dai, and X. Huang, Pre-trained models for natural language processing: A survey, *Sci. China Technol. Sci.*, vol. 63, no. 10, pp. 1872–1897, 2020.
[42] H. Wang, Y. Wang, X. Song, B. Zhou, X. Zhao, and F. Xie, Quantifying controversy from stance, sentiment,

Zixuan Wang is a PhD candidate at School of Computer Software, Tianjin University, China. He received the BEng degree from University of Electronic Science and Technology, China in 2016, and the MEng degree from Tianjin University, China in 2022. His research interests include natural language processing and large language models.

Zhiyong Feng is a professor at School of Computer Software, Tianjin University, China. He received the BEng degree from Inner Mongolia Agricultural University, China in 1985, the MEng degree from Jilin Polytechnical University, China in 1988, and the PhD degree from Tianjin University, China in 1996. His research interests include artificial intelligence and service computing.

Yan Zeng is an undergraduate student at College of Life Sciences, Tianjin University, China. Her main research interest is intelligent prediction of virus structure.

Rafiul Haq is a PhD candidate at School of Computer Software, Tianjin University, China. He received the BEng degree from University of Malakand, Pakistan, and the MEng degree from COMSATS University Islamabad, Pakistan. His research interests include natural language processing, machine learning, deep learning, and data mining, focusing on sequence labeling tasks in low-resource languages such as Urdu.

Xiping Zhao is a master student at College of Mechanical Engineering, Tianjin University, China. He received the BEng degree from Tianjin University, China in 2025. His main research interest is the design of intelligent energy management systems.

Kaibing Yang is an undergraduate at College of Future Technology, Tianjin University, China. His research interests include multimodal perception, generation, and reasoning.

Wenqi Cao is a master student at College of Mechanical Engineering, Tianjin University, China. He received the BEng degree from Tianjin University, China in 2025. His main research interest is intelligent recognition in vision.

offensiveness and sarcasm: A fine-grained controversy intensity measurement framework on a Chinese dataset, *World Wide Web*, vol. 26, no. 5, pp. 3607–3632, 2023.

[43] B. Liang, Z. Lin, R. Xu, and B. Qin, Topic-oriented sarcasm detection: New task, new dataset and new method, (in Chinese), *J. Chin. Inf. Process.*, vol. 37, no. 2, pp. 138–147,157, 2023.